

Class 09

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2. Importing Candy Data

```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

2.1 What is in this data set?

Q1. How many different candy types are in this dataset?

There are 85 different candies.

```
dim(candy)
```

```
[1] 85 12
```

```
nrow(candy)
```

```
[1] 85
```

```
sum(candy$fruity)
```

```
[1] 38
```

Q2. How many fruity candy types are in the dataset?

There are 38 fruity candy types in this data set.

2.2 What is your favorite candy?

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

The candy I chose other than Twix is Almond Joy. The winpercent value is 50.3475.

```
candy["Almond Joy", ]$winpercent
```

```
[1] 50.34755
```

Q4. What is the winpercent value for “Kit Kat”?

The winpercent value for Kit Kat is 76.7686.

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

The winpercent value for tootsie roll snack bars is 49.6535.

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

side note: the skimr::skim() function

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

winpercent seems to be on a different scale. It looks like a percentage as opposed to a proportion.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

I think it's a binary for yes and no. 0 is likely no and 1 is likely yes.

```
skim(candy$chocolate)
```

Table 3: Data summary

Name	candy\$chocolate
Number of rows	85
Number of columns	1
Column type frequency: numeric	1
Group variables	None

Variable type: numeric

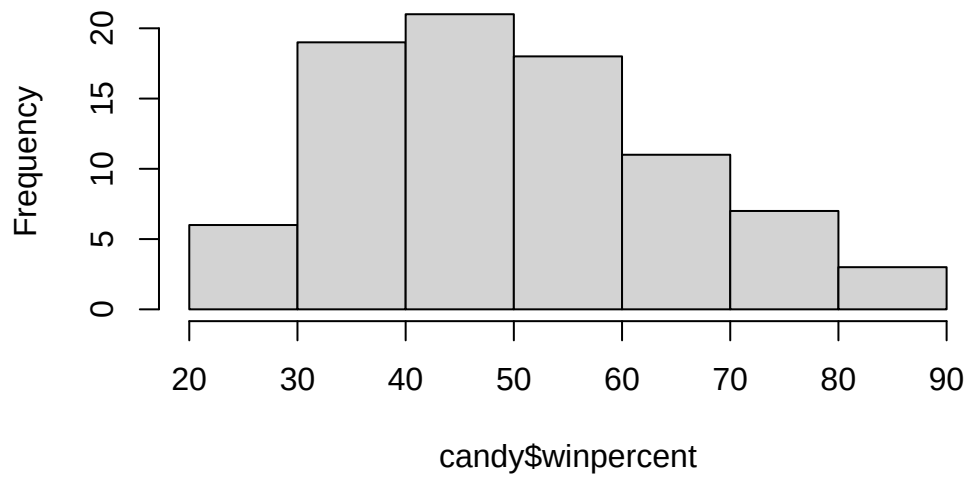
skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
data	0	1	0.44	0.5	0	0	0	1	1	

3 Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

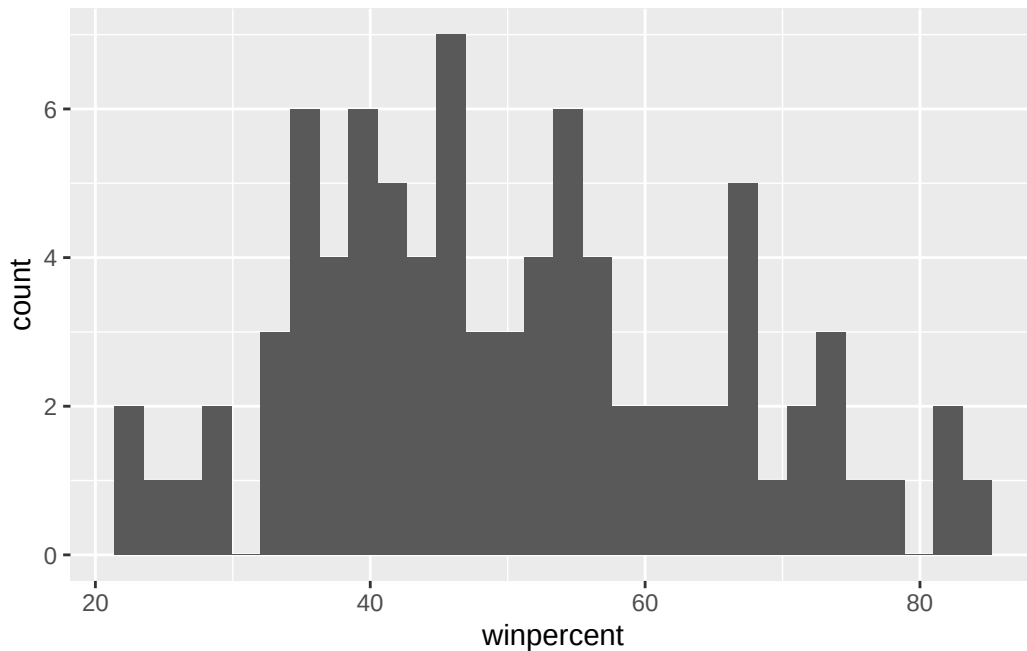
```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
library(ggplot2)
ggplot(candy, aes(x=winpercent)) +
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Q9. Is the distribution of winpercent values symmetrical?

It does not look very symmetrical. It looks skewed to the right.

Q10. Is the center of the distribution above or below 50%?

```
median(candy$winpercent)
```

```
[1] 47.82975
```

The center of the distribution is below 50% at 47.83%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
fruity_candy <- candy$winpercent[as.logical(candy$fruity)]
```

```
chocolate_candy <- candy$winpercent[as.logical(candy$chocolate)]
```

```
mean(fruity_candy)
```

```
[1] 44.11974
```

```
mean(chocolate_candy)
```

```
[1] 60.92153
```

On average, chocolate candy is more highly ranked (60.92%) than fruity candy (44.12%).

Q12. Is this difference statistically significant?

```
t.test(chocolate_candy, fruity_candy)
```

Welch Two Sample t-test

```
data: chocolate_candy and fruity_candy
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

The difference is statistically significant as the p value is less than or equal to .05 (p=2.871e-08).

4 Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

The five least liked candy types in this set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```
candy |> arrange(winpercent) |> head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0

Jawbusters	0	1	0	0	0	
	crisped	ricewafer	hard bar	pluribus	sugarpercent	pricepercent
Nik L Nip		0	0	0	1	0.197 0.976
Boston Baked Beans		0	0	0	1	0.313 0.511
Chiclets		0	0	0	1	0.046 0.325
Super Bubble		0	0	0	0	0.162 0.116
Jawbusters		0	1	0	1	0.093 0.511
	winpercent					
Nik L Nip	22.44534					
Boston Baked Beans	23.41782					
Chiclets	24.52499					
Super Bubble	27.30386					
Jawbusters	28.12744					

Q14. What are the top 5 all time favorite candy types out of this set?

The 5 all time favorite candies out of this data set are Snickers, Kit Kat, Twix, Reese's Miniatures, and Reese's Peanut Butter cup.

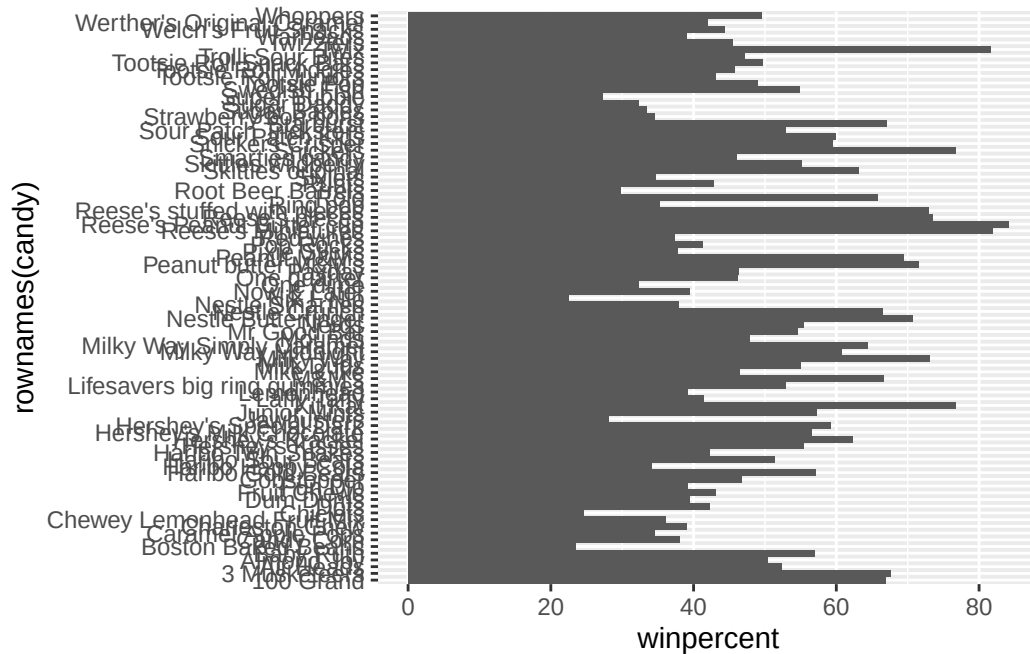
```
candy |> arrange(winpercent) |> tail(5)
```

	chocolate	fruity	caramel	peanut	almondy	nougat
Snickers	1	0	1		1	1
Kit Kat	1	0	0		0	0
Twix	1	0	1		0	0
Reese's Miniatures	1	0	0		1	0
Reese's Peanut Butter cup	1	0	0		1	0
	crisped	ricewafer	hard bar	pluribus	sugarpercent	
Snickers		0	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Twix		1	0	1	0	0.546
Reese's Miniatures		0	0	0	0	0.034
Reese's Peanut Butter cup		0	0	0	0	0.720
	pricepercent	winpercent				
Snickers	0.651	76.67378				
Kit Kat	0.511	76.76860				
Twix	0.906	81.64291				
Reese's Miniatures	0.279	81.86626				
Reese's Peanut Butter cup	0.651	84.18029				

Q15. Make a first barplot of candy ranking based on winpercent values.

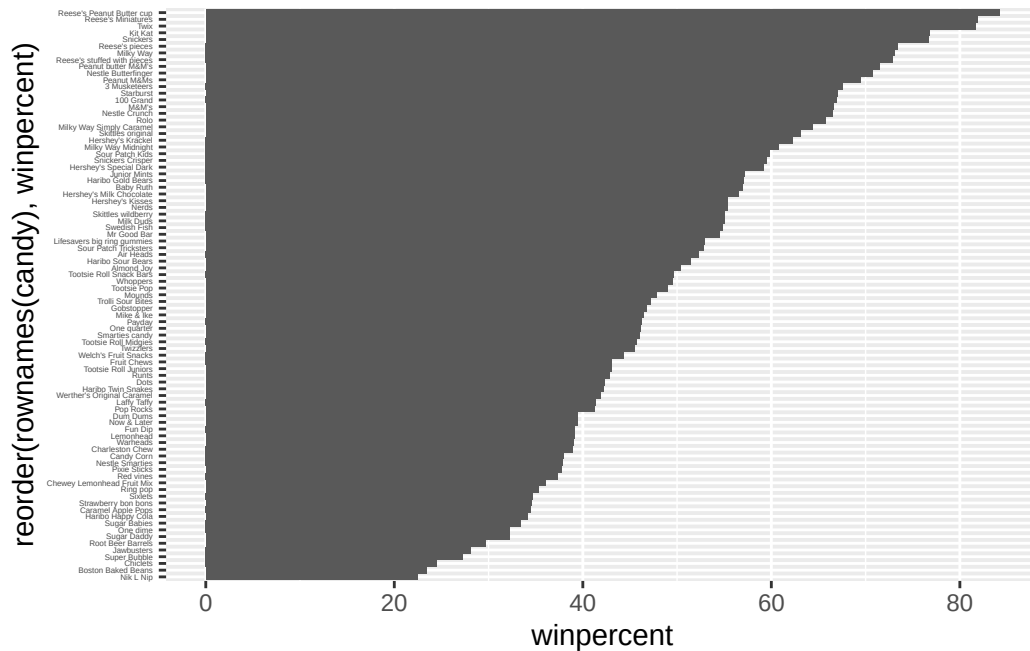
```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

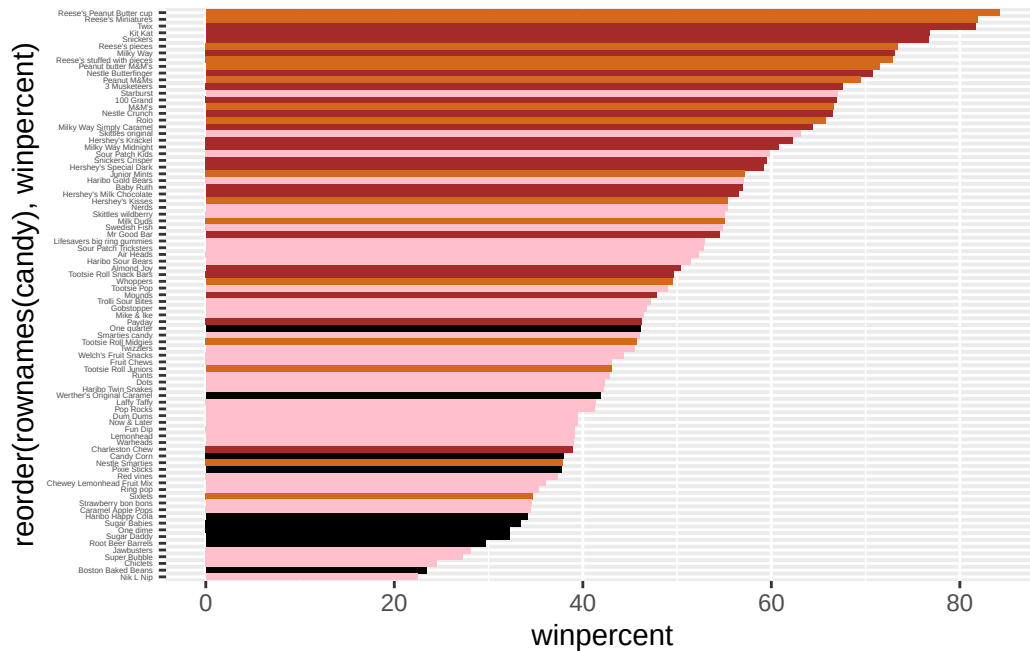
```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col() +
  theme(axis.text.y = element_text(size = 3))
```



4.1 Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  theme(axis.text.y = element_text(size = 3))
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy are sixlets. > Q18. What is the best ranked fruity candy?

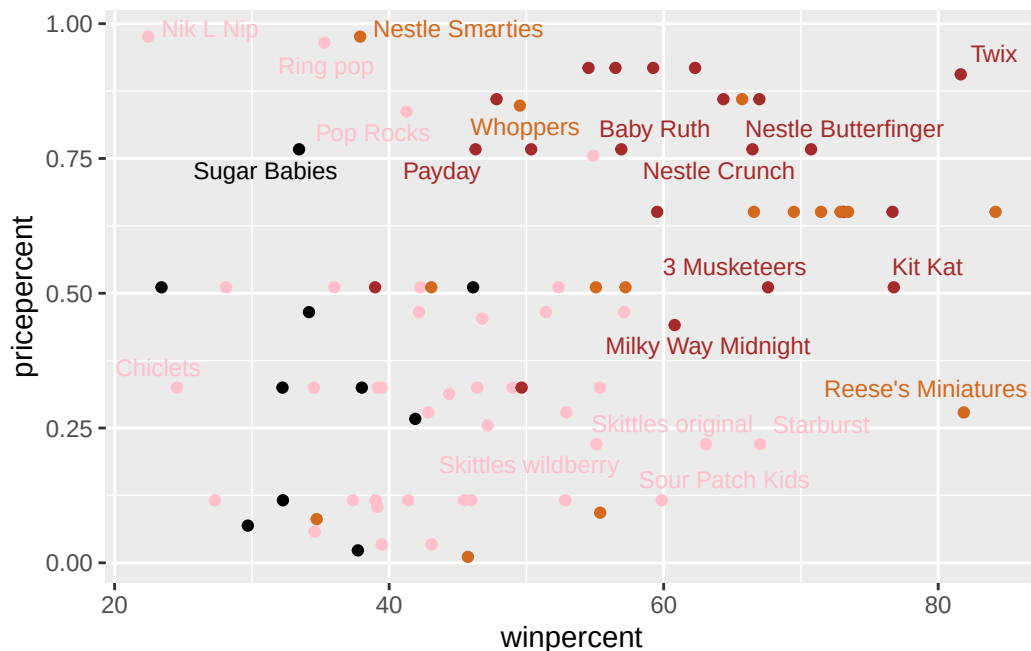
The best ranked fruity candy are starburst.

5 Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Tootsie Roll Midgies have the highest winpercent for the lowest pricepercent.

```
best <- candy$winpercent / candy$pricepercent
candy[which.max(best), c("winpercent", "pricepercent")]
```

	winpercent	pricepercent
Tootsie Roll Midgies	45.73675	0.011

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The tip 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. Of these the least popular is Nik L Nip.

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

pricepercent	winpercent
--------------	------------

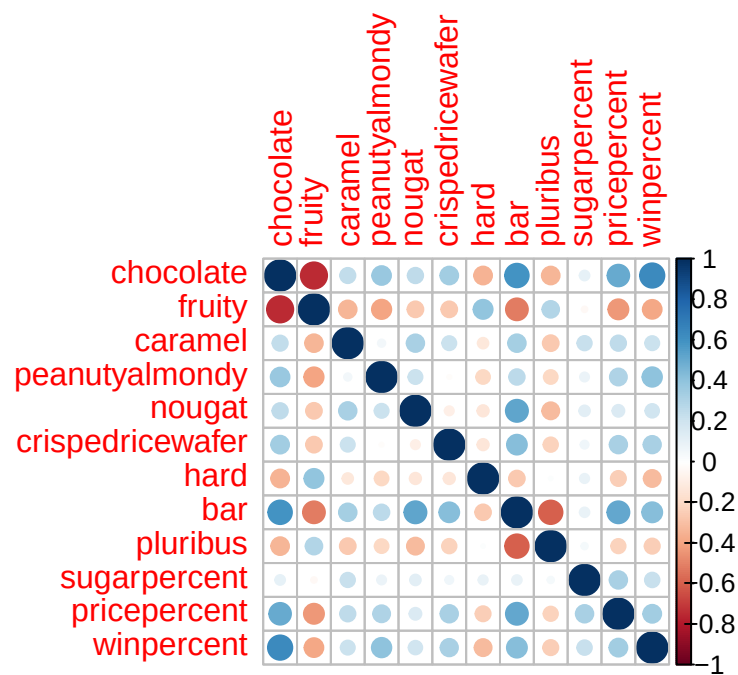
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

6 Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

It seems like fruity and a variety of things. Of note are fruity and chocolate, fruity and bar, and pluribus and bar (among others). > Q23. Similarly, what two variables are most positively correlated?

It looks like it would be chocolate and bar. It seems many chocolate + XX pairs are blue (positively correlated).

7 Principal Component Analysis

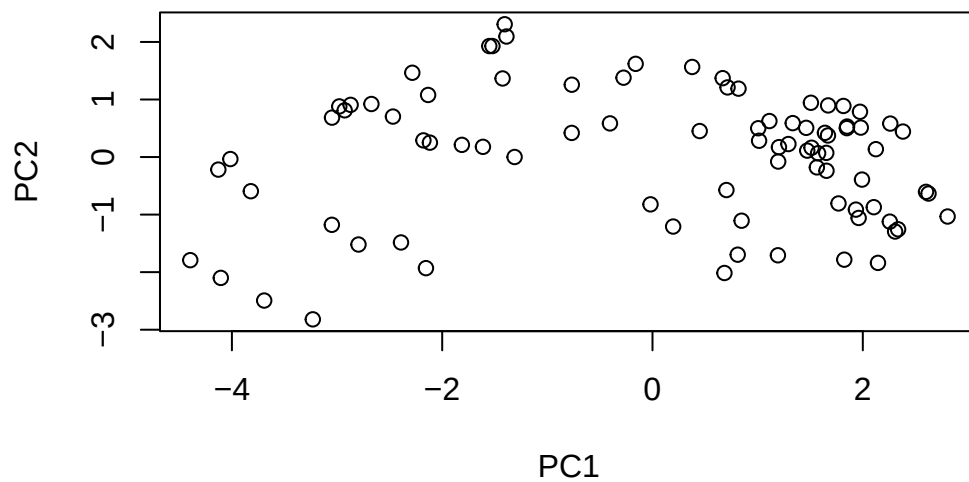
```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

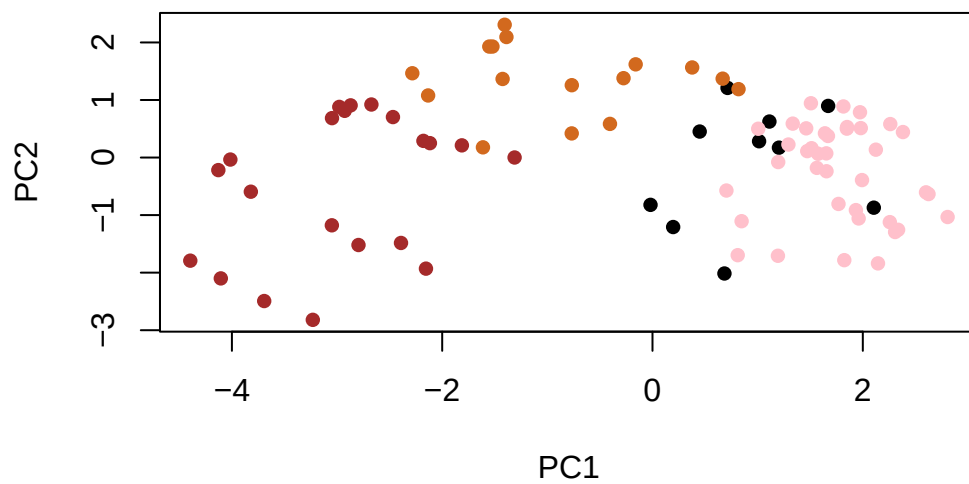
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2])
```



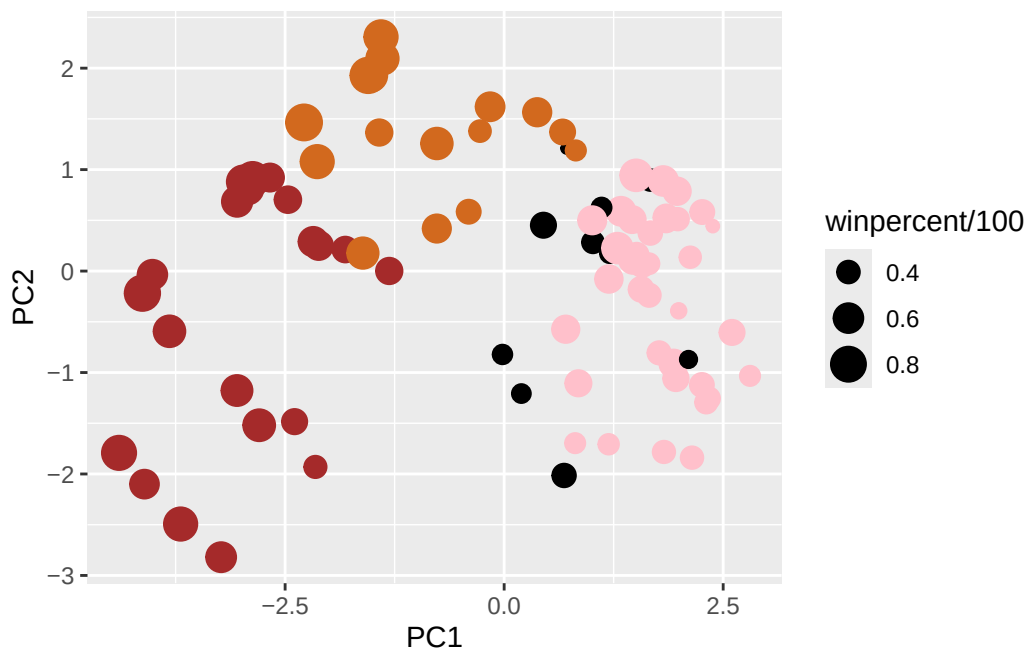
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
```

p



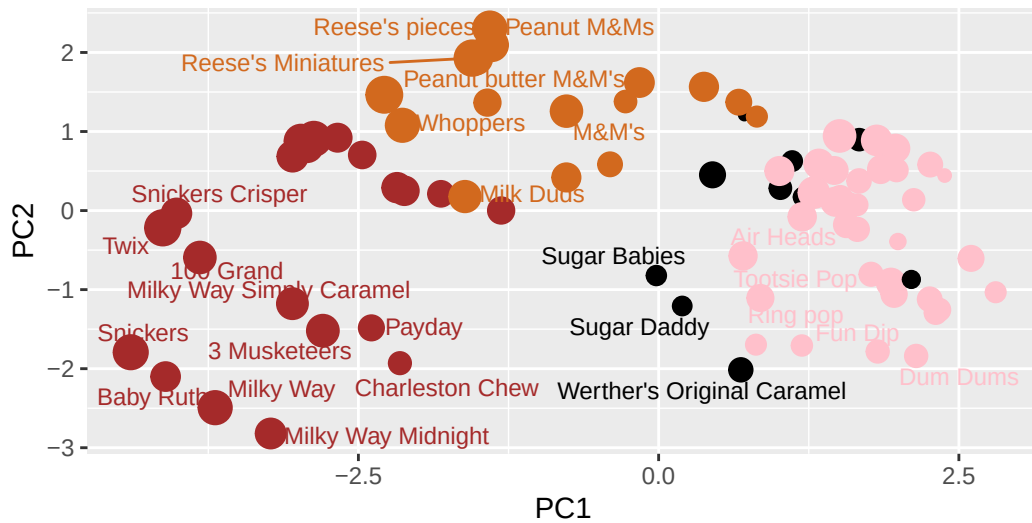
```
library(ggrepel)
```

```
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
       caption="Data from 538")
```

Warning: ggrepel: 59 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

```
last_plot
```

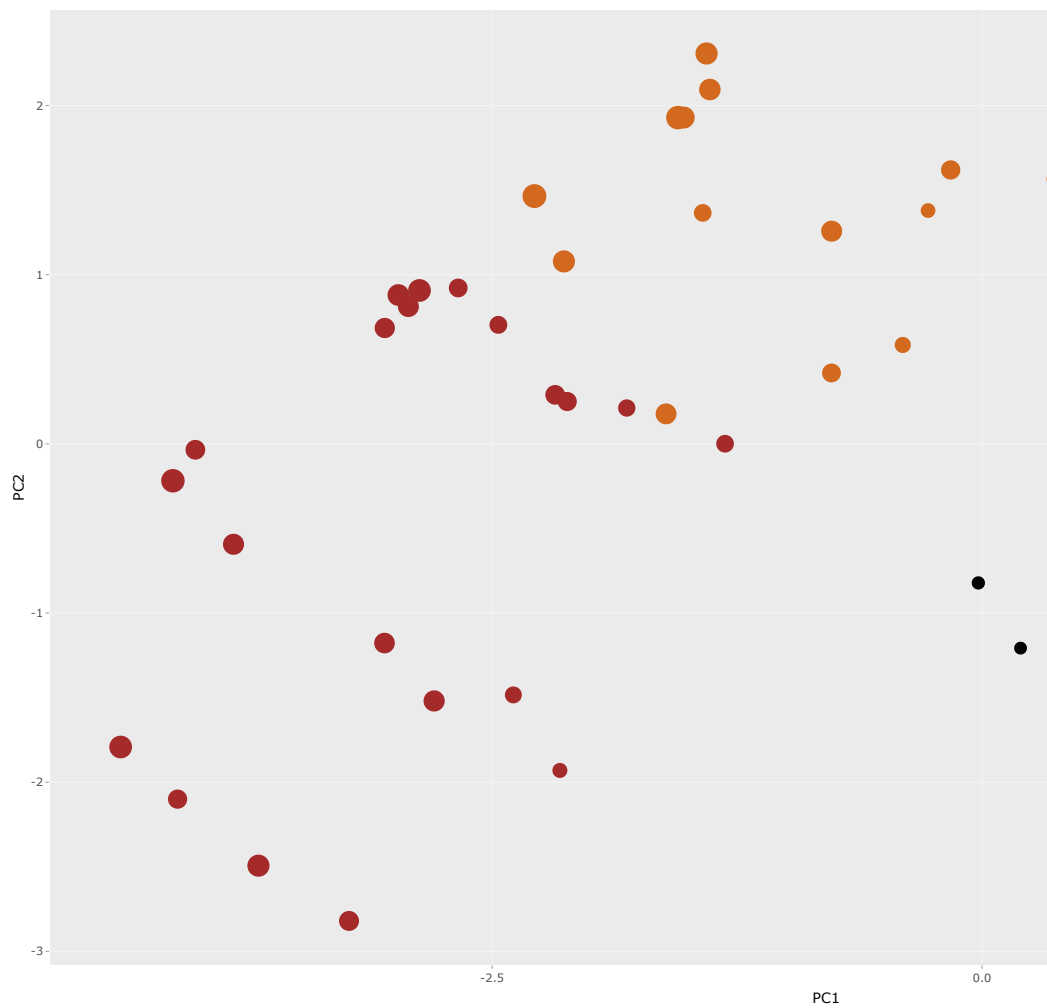
The following object is masked from 'package:stats':

```
filter
```

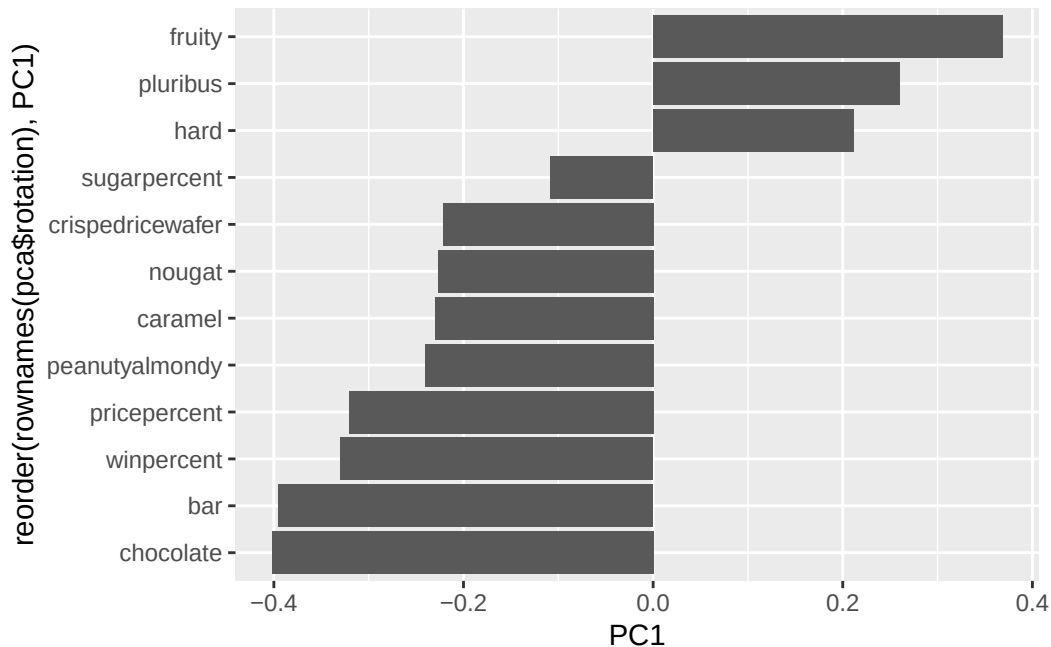
The following object is masked from 'package:graphics':

```
layout
```

```
ggplotly(p)
```



```
ggplot(pca$rotation) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

In the positive direction, fruity, pluribus, and hard are picked up. It feels a little confusing. They were negatively/ anticorrelated previously.

8 Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The winning candy has chocolate, nougat, bars, and peanutyalmondy. The visualizations allowed us to look at costs and like-ability. The correlations helped us see the best combinations that led to better liking. PCA provided predictive powers.